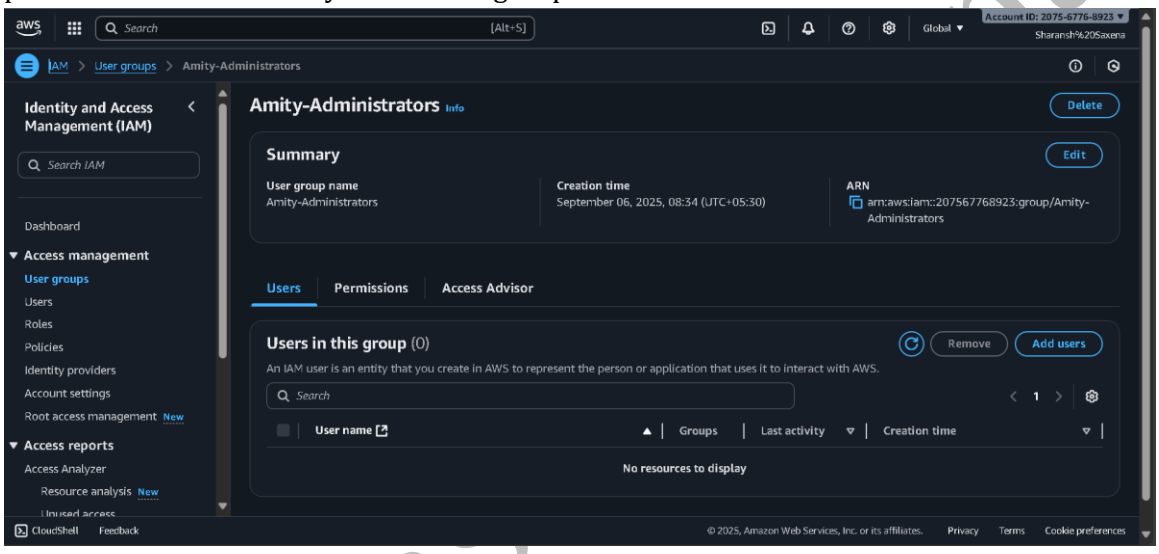
Applied Cloud Computing – DIY Project 1

This document contains the step-by-step execution of AWS IAM and EC2 tasks as part of DIY Project 1 for Applied Cloud Computing. Each task includes explanation, commands and screenshots.

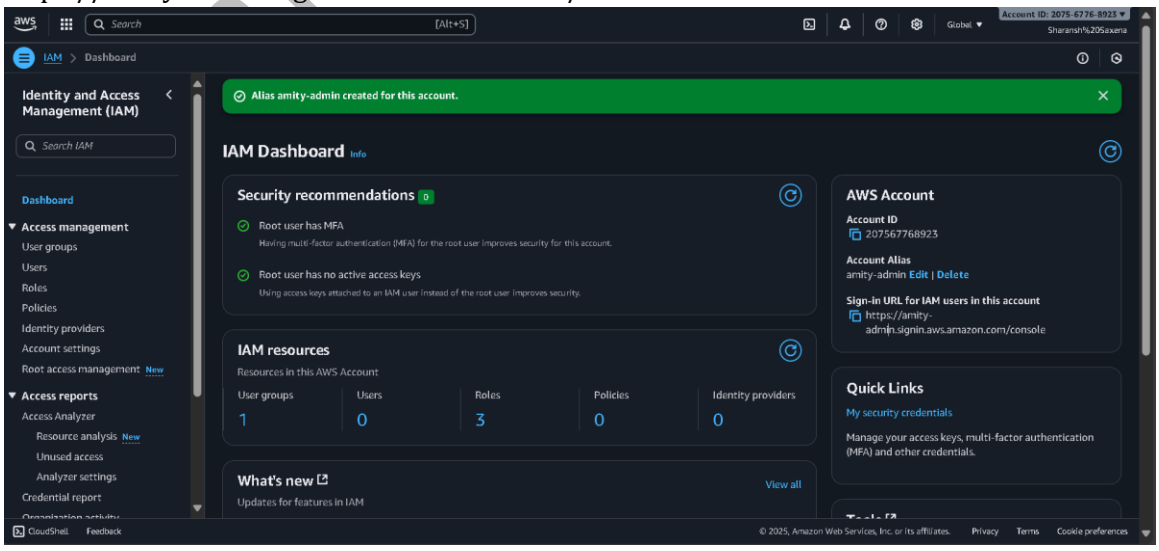
Step 1: Create IAM Group

Login as root user and create an IAM group called Administrators. Attach the managed policy 'Amity-Administrators' to this group. Since AWS reserves the name **Administrator1** for internal use, I created a group named **Amity-Administrators** with the same permissions and added my user to this group.



Step 2: Customize Sign-in Link

Customize a sign-in link for your AWS account and note the new link. Example: https://amity-admin.signin.aws.amazon.com/console



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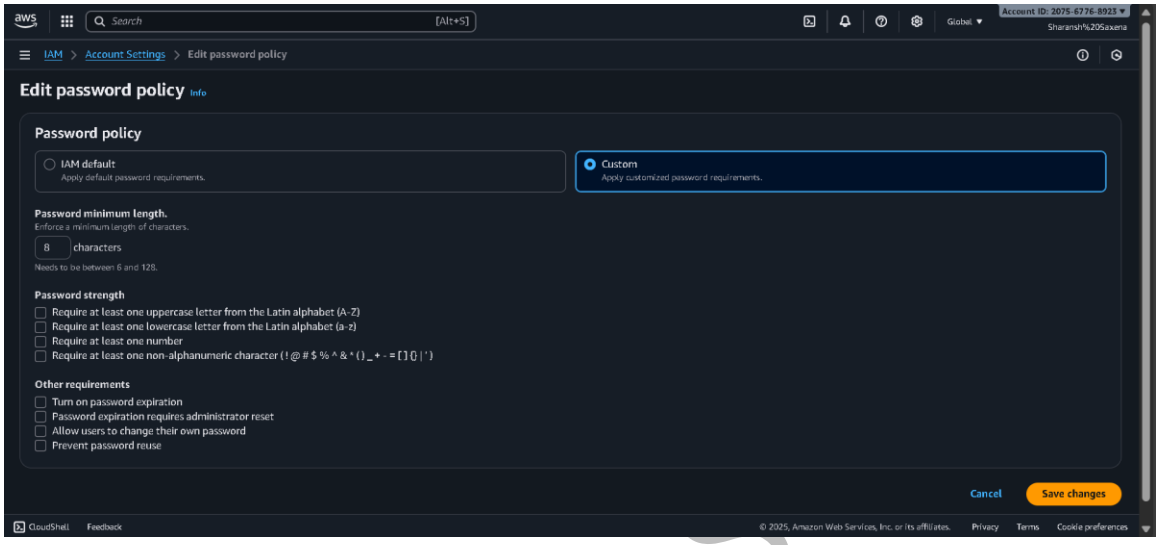
Step 3: Create Password Policy

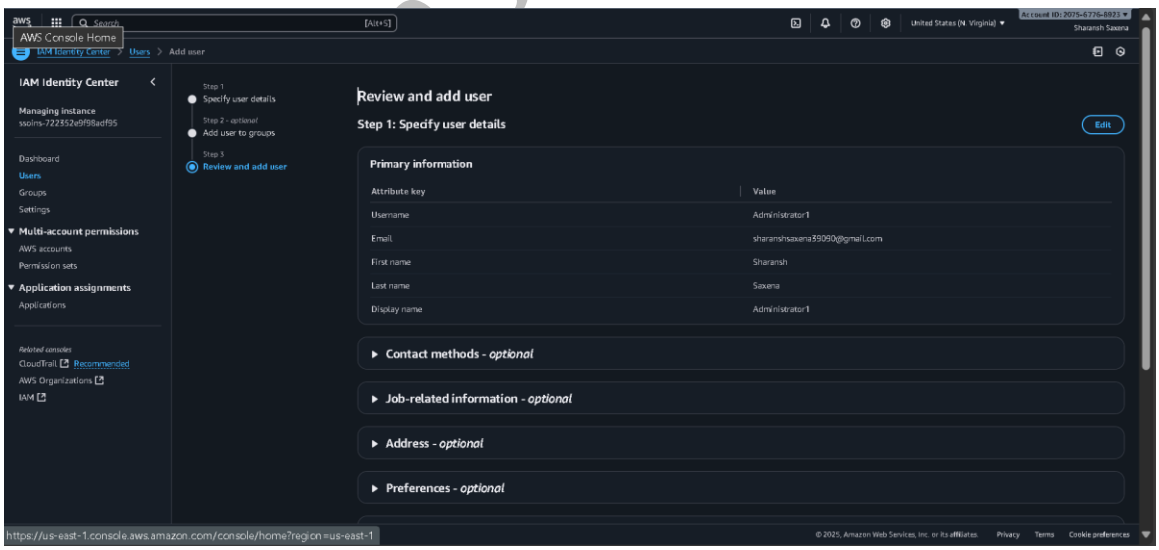
Create a strong password policy for your AWS account. Set requirements such as minimum length, uppercase, numbers, and symbols.



Step 4: Create IAM User

While logged in as root user, create a new IAM user called 'Administrator' and add this user to the Administrators group.

Since the Review page could not fit in one screenshot, I have captured it in two parts.



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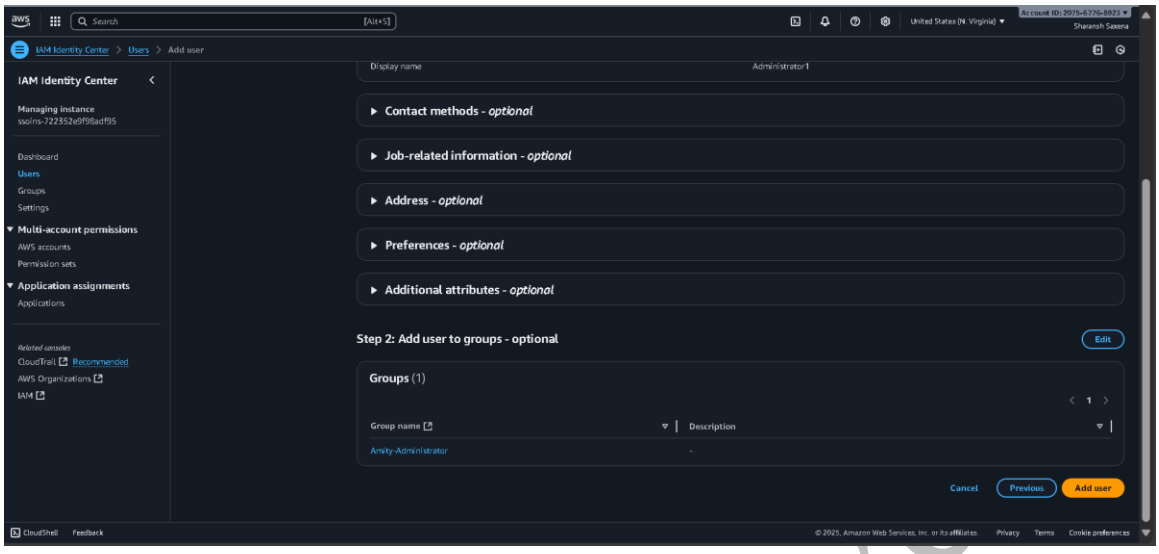

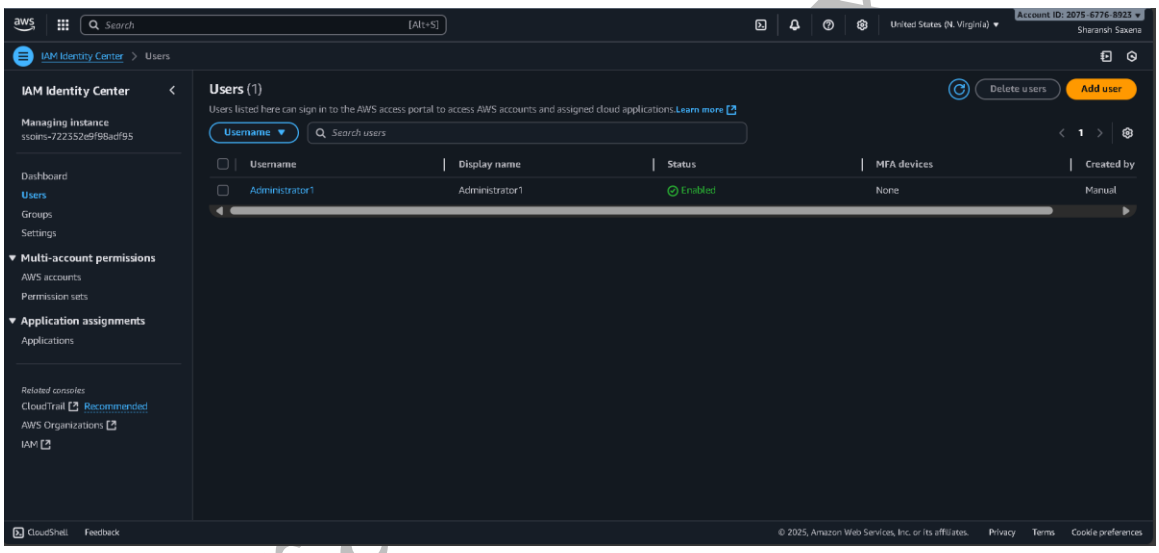

Users list showing Administrator1 enabled and visible.

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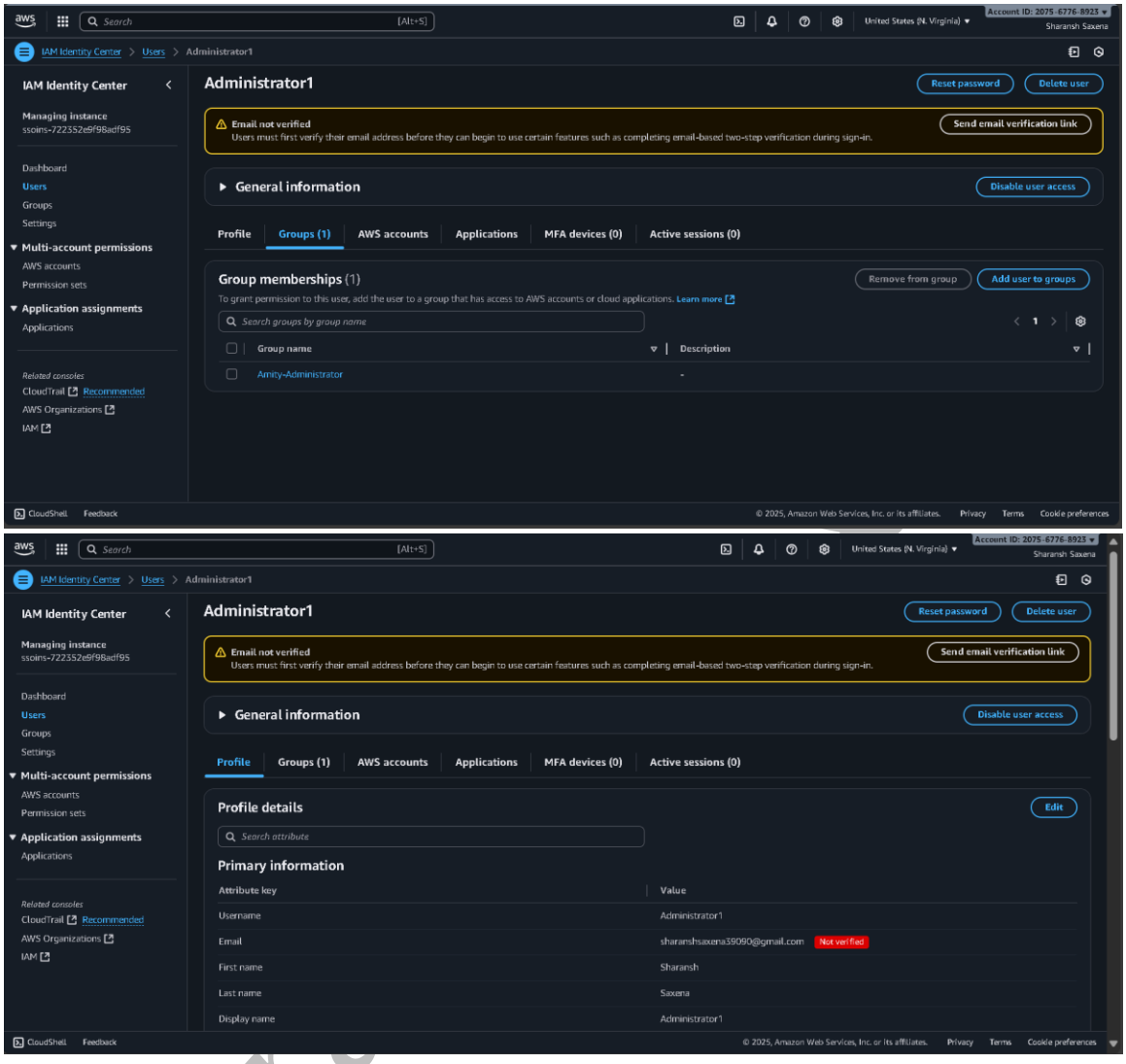

User details page of Administrator1, showing Group memberships section with Amity-Administrators.

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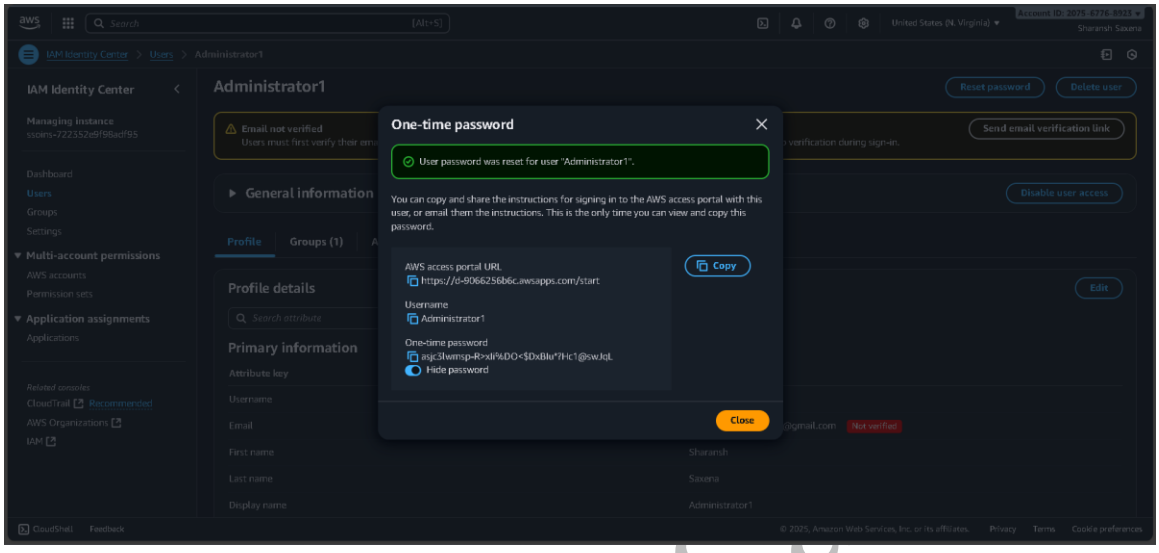
Step 5: Create Password for IAM User

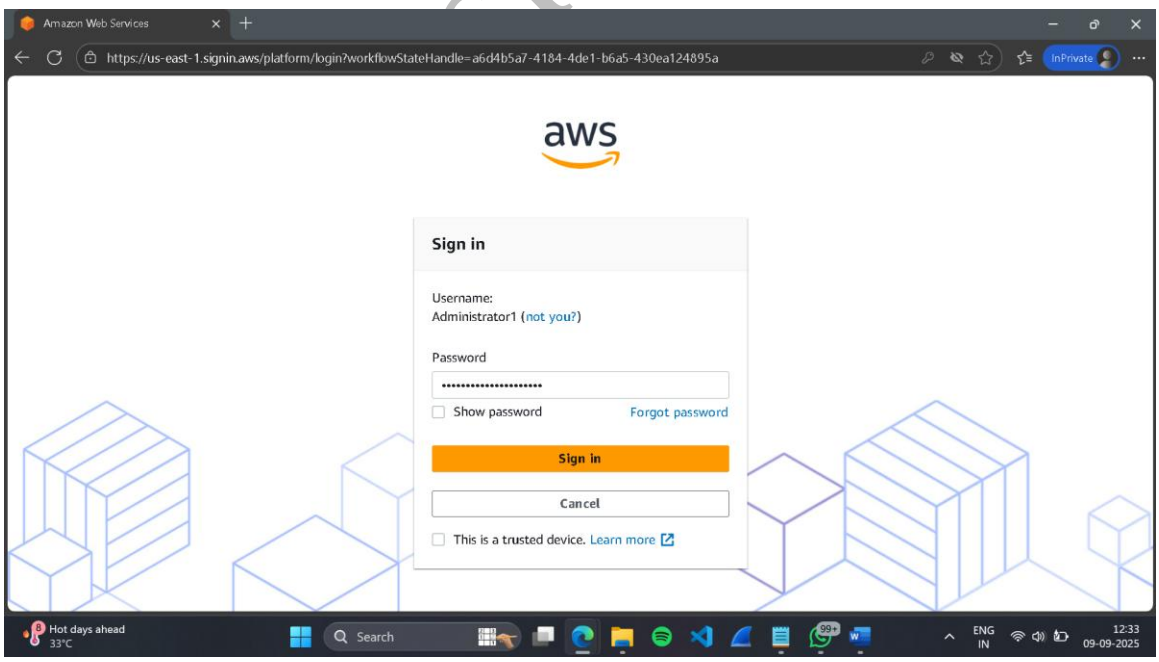
On the details page for the Administrator user, set a password for console access.



Confirmation page showing AWS Access Portal URL, Username (Administrator1), and the generated one-time password.

Step 6: Login as Administrator

Log out as root user and log in with the customized sign-in link using the Administrator user credentials.



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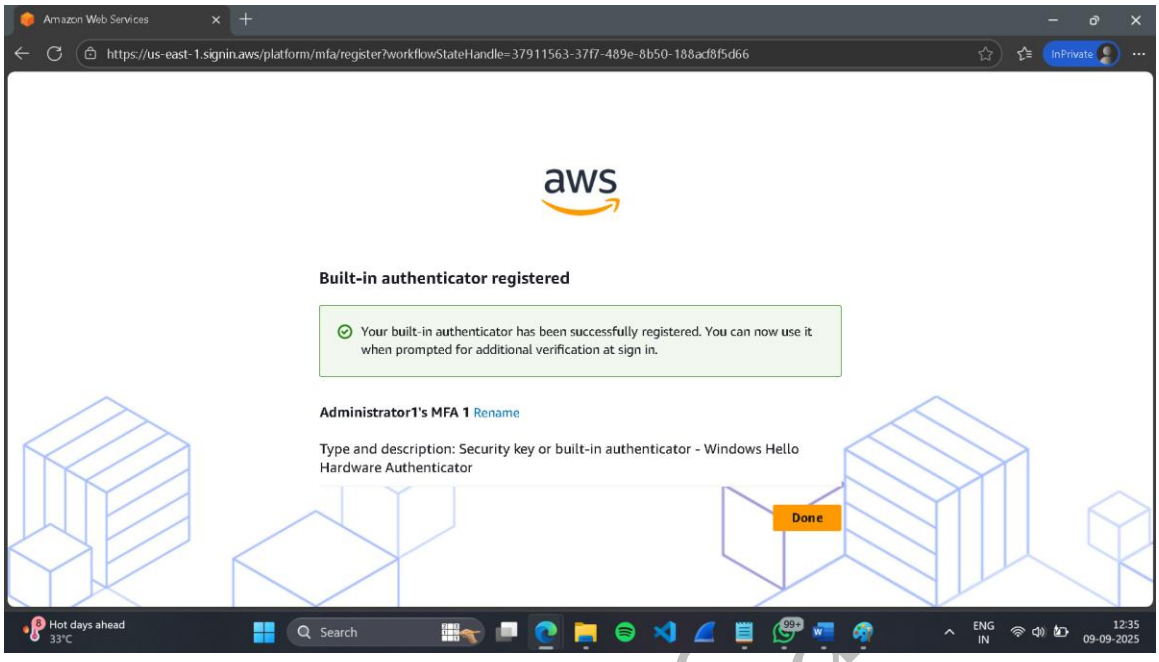

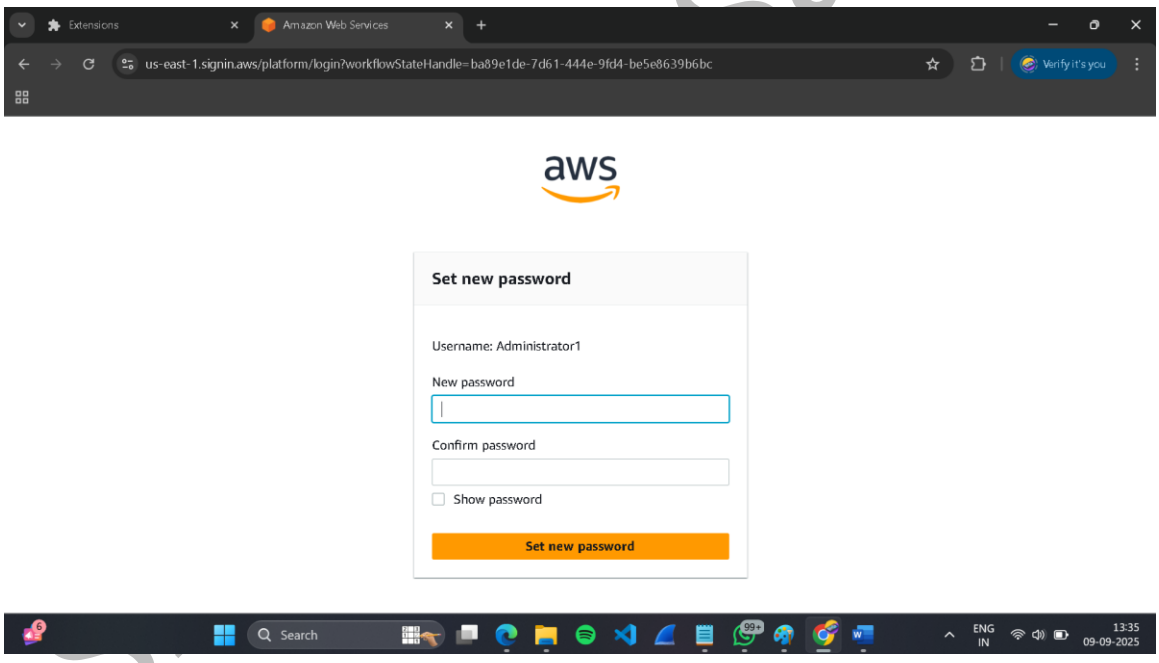

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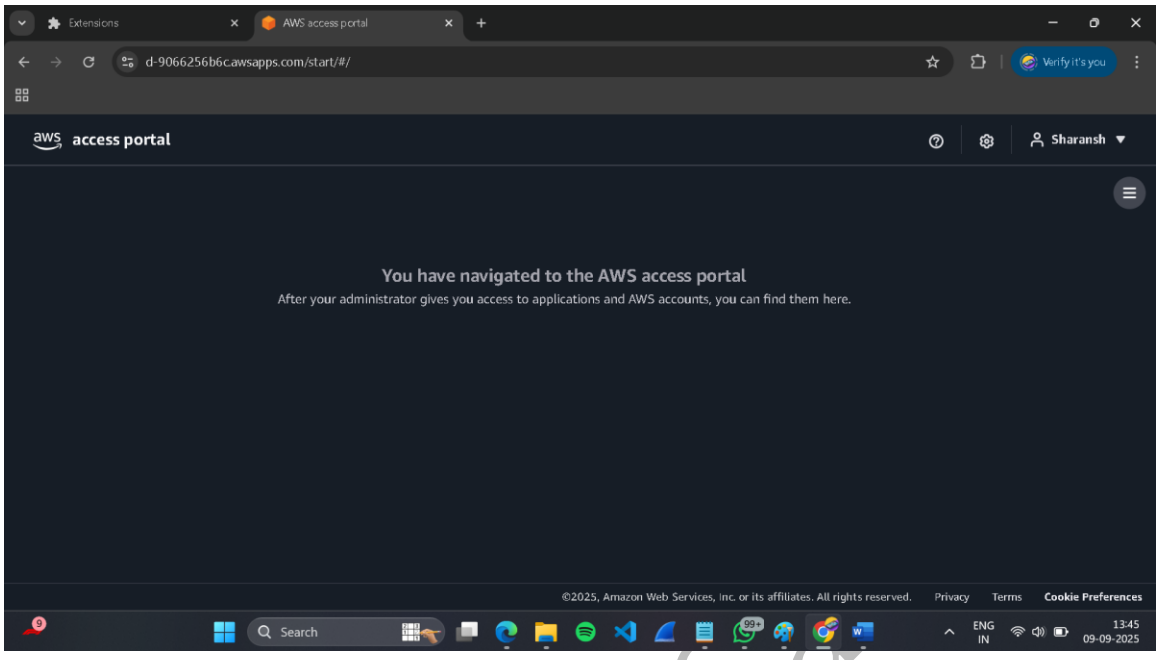
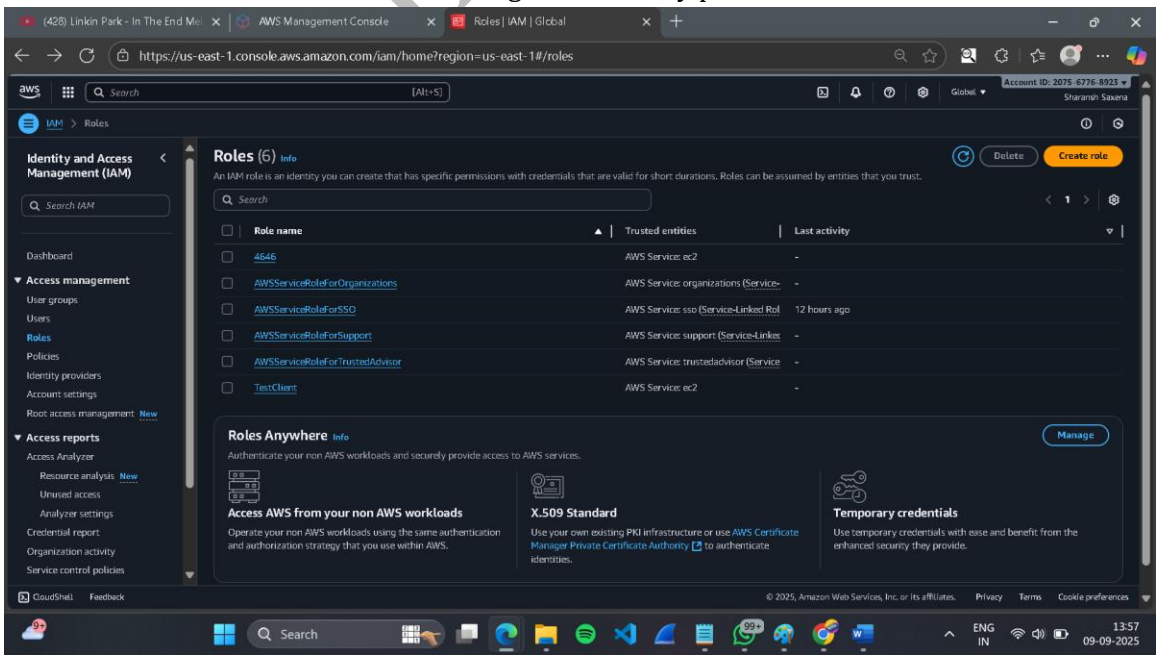


The above screenshot shows the successful login of the newly created user Administrator1 through the customized AWS Access Portal sign-in link. After entering the one-time password and resetting it to a new strong password, the user was able to access the AWS Access Portal dashboard.

Step 7: Create IAM Role for EC2

While signed in as the Administrator, create an IAM role named **TestClient** with trusted entity type **EC2**. Attach the AWS managed policy **ViewOnlyAccess** to this role. This allows the EC2 instance to assume the role and gain view-only permissions for AWS resources.



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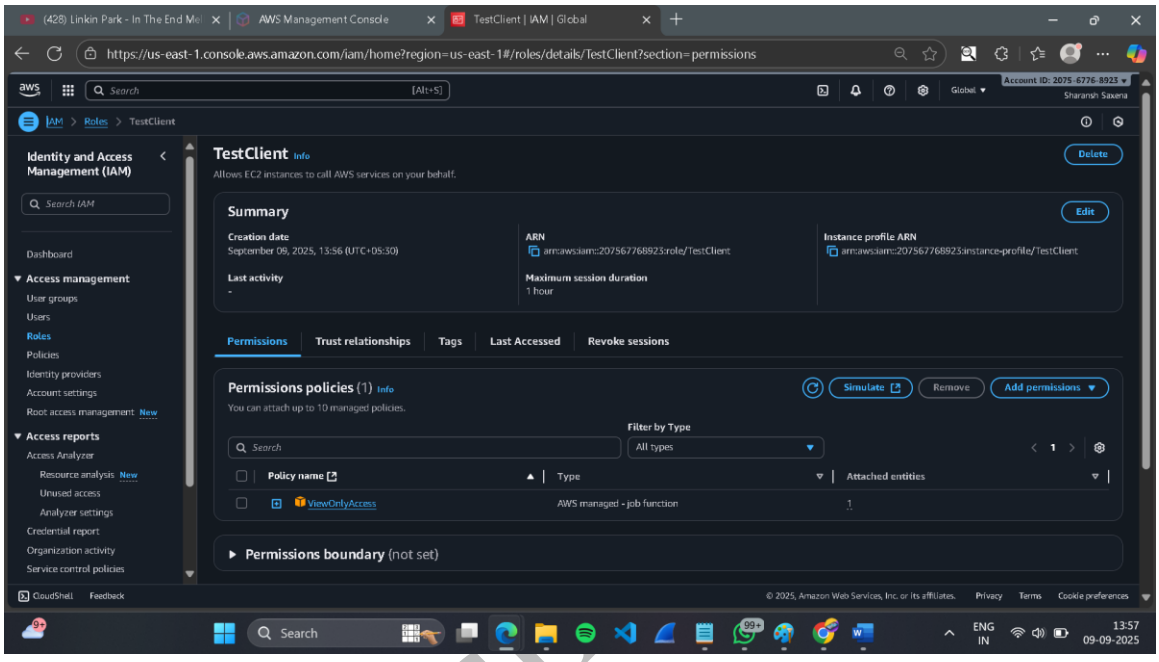

The IAM role “TestClient” was created with EC2 as the trusted entity. Since the “ReadOnlyAccess” policy was not available in my AWS account, I attached the AWS managed policy “ViewOnlyAccess.” This policy provides equivalent view-only permissions to AWS resources and fulfills the assignment requirement.

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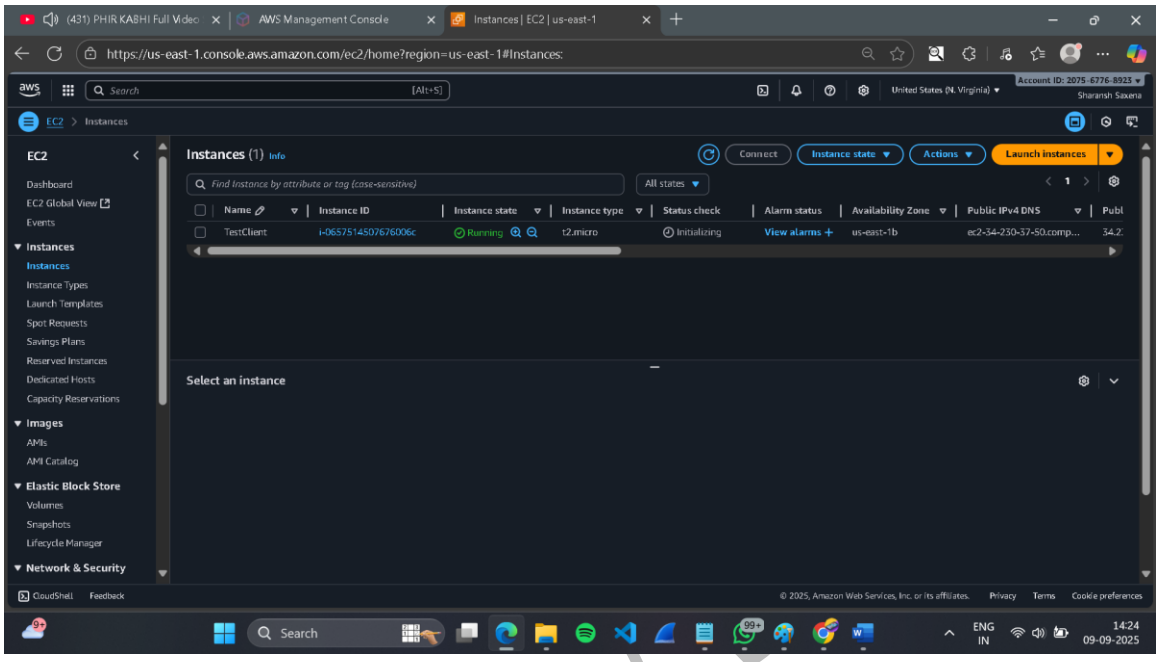
Step 8: Launch EC2 Instance

Launch an Amazon Linux EC2 instance and attach the 'TestClient' role to it.



The above screenshot shows the EC2 instance "TestClient (Instance ID: i-0657514507676006c)" running in Availability Zone us-east-1b. The instance is of type t2.micro, launched with the key pair "TestClient-Key" and attached to the IAM role "TestClient". The public IPv4 address 34.230.37.50 will be used to connect via SSH.

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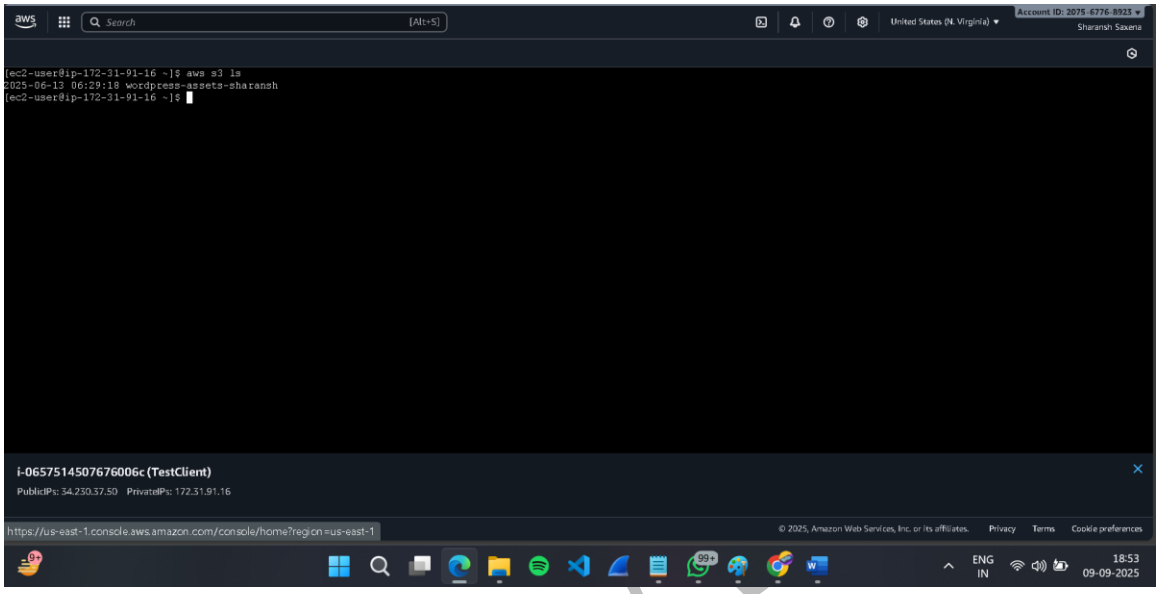
Step 9: SSH into EC2 & List S3 Buckets

SSH into the EC2 instance and run the following command to list S3 buckets:

aws s3 ls



The above screenshot shows the EC2 instance successfully accessing Amazon S3 using the IAM role TestClient. The command aws s3 ls lists the available bucket wordpress-assets-sharansh, verifying that the role permissions are working correctly.

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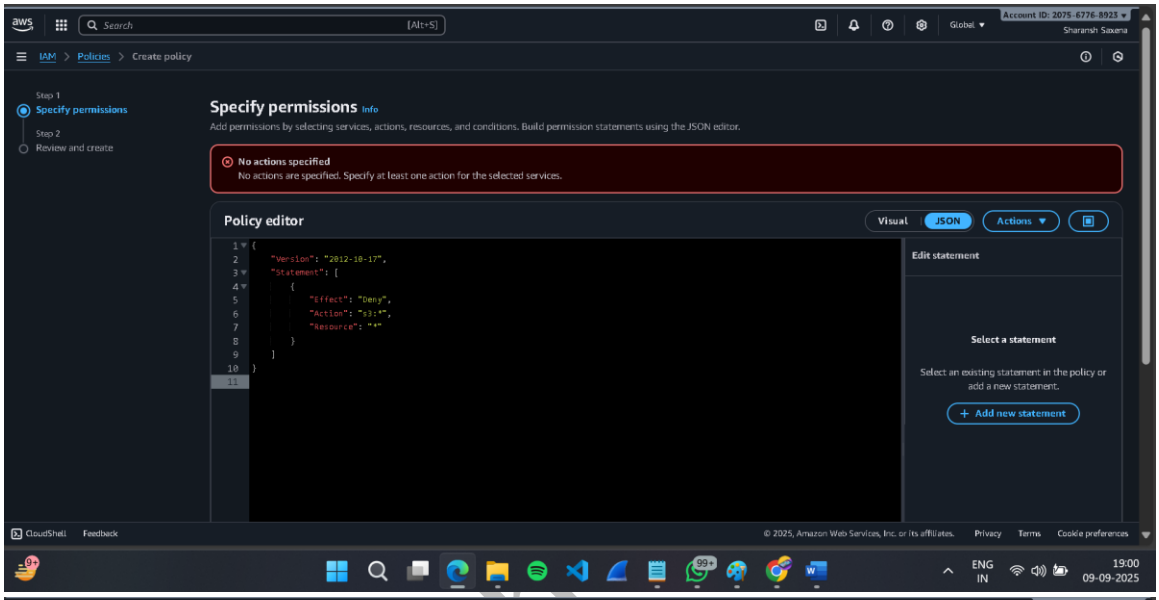
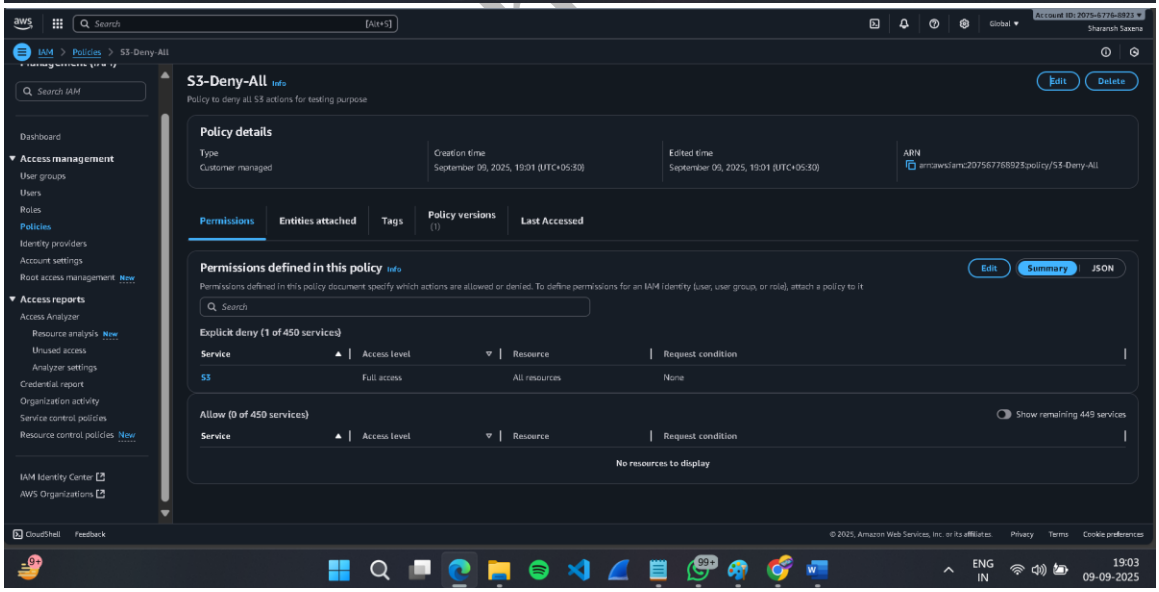
Step 10: Create Conflicting Policy

Use the policy generator to create a new policy with the following details:
- Effect: Deny
- Service: Amazon S3
- Actions: *
- ARN: *

Attach this policy to the Administrators group.





The above screenshot shows the custom IAM policy S3-Deny-All created to explicitly deny all Amazon S3 actions across all resources. This policy is customer managed and will be attached to the Amity-Administrators group to test the Deny precedence over Allow policies.

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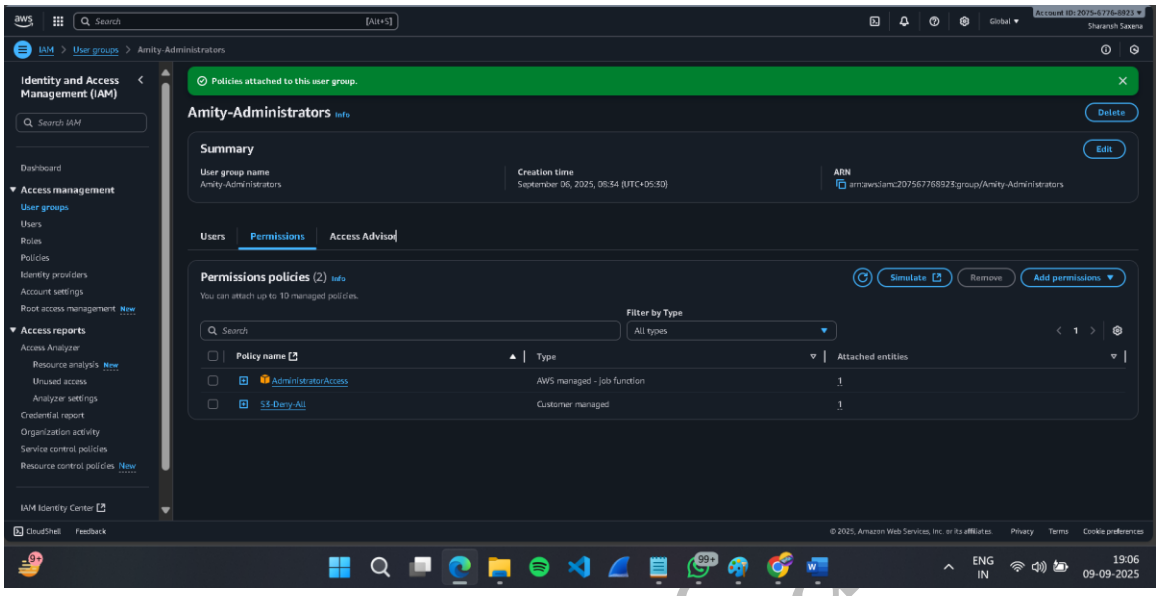

The above screenshot shows the custom IAM policy S3-Deny-All created using the JSON editor. This policy explicitly denies all Amazon S3 actions across all resources and will later be attached to the Amity-Administrators group to test Deny precedence.

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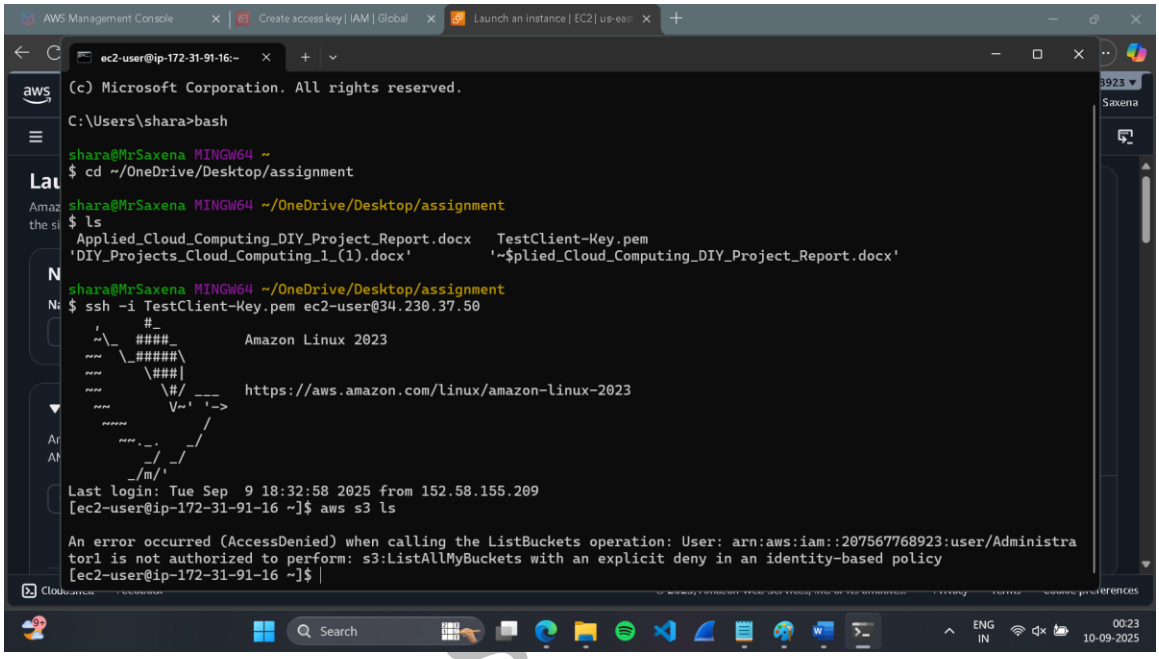
Step 11: Test Access Denial

Use the AWS CLI again to try listing S3 buckets. Even though the 'ReadOnlyAccess' policy allows access, the new Deny policy overrides it, so access is denied.

aws s3 ls



The above screenshot shows the EC2 instance user Administrator1 attempting to list S3 buckets. Due to the attached S3-Deny-All policy, the operation was denied with an explicit AccessDenied error. This validates that Deny policies override any Allow permissions in AWS IAM.